

Motivation with a reference to existing work

Inspired by BabyLM challenge, I want to explore whether language models trained on a small dataset are better suited than larger ones for helping adults learn a foreign language, particularly in terms of grammar. Since LLMs often completely rewrite sentences written by non-native speakers to match the style of native speakers. In contrast, BabyLM has lower cognitive complexity and may more naturally align with the pedagogical principle of “minimal modification,” thereby avoiding “overcorrection.”

For learners, using BabyLM offers some advantages, such as faster response and lower computational costs (or even no cost at all). If small models perform better at “moderate error correction,” this will greatly advance the widespread adoption of low-cost educational AI.

Reference:

Warstadt et al (2023) <https://www.research-collection.ethz.ch/server/api/core/bitstreams/419b7c31-5307-461f-811b-5241b4fc26ca/content> “Findings of the BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora”

Staruch et al (2025) <https://arxiv.org/pdf/2506.13148> “Adapting LLMs for Minimal-edit Grammatical Error Correction”

Fatemi et al (2025) https://home.citl.labs.vu.nl/static/data/theses/thesis_Farnaz_Bani_Fatemi_2025.pdf “Grammaticality and LLMs: Evaluating the Potential of BabyLMs for Grammatical Error Detection in NLP”

research question

When correcting grammatical errors made by second-language beginners (e.g., at the A1/A2 levels), do BabyLM exhibit a lower tendency to “overcorrection” than large language models?

Plan to answer the research question

Select a high-performing BabyLM and a widely used large language model as a control group.

Use publicly available error-correction corpora for second-language learners (e.g., the BEA-2019 Shared Task dataset) to extract learners’ original sentences containing typical grammatical errors.

1. Prompting: Design prompts instructing both models to correct grammatical errors in the learners’ sentences.
2. Metric 1 (Overcorrection Rate): Calculate the Levenshtein distance between the corrected sentence generated by the model and the learner’s original sentence. Assuming both models successfully correct the grammatical error, a smaller Levenshtein distance indicates a smaller scope of modification, which better aligns with the principle of “avoiding overcorrection.”
3. Metric 2 (Lexical Difficulty Penalty): Determine whether the “new words” introduced by the model exceed the basic vocabulary list (based on word frequency statistics for A1/A2 levels) to quantify whether the model has introduced unnecessary advanced vocabulary.

